

ePC-SAFT For Produced-Water Lithium Extraction

Thermodynamic and process-systems case study for HBTA/TOPO solvent extraction

Chemical engineering technical review

2026-05-07

0. Technical Thesis

This deck makes one chemical-engineering argument:

Produced-water lithium extraction needs a thermodynamic model that maps basin chemistry into solvent-extraction transfer variables. The benchmark chemistry is the non-ionic HBTA/TOPO/sulfonated-kerosene system; ePC-SAFT is evaluated as the equilibrium backbone for PrOMMiS and IDAES.

- Feed basis: southern Arkansas Smackover brine.
- Solvent chemistry: HBTA/TOPO complexation with selective Li transfer.
- Thermodynamic target: electrolyte liquid-liquid equilibrium and ion-specific distribution.
- Process target: staged extraction, surrogate generation, and IDAES costing.

1. Basin Thermodynamic Basis

1.1 Produced Water Is Not One Feed

Basin	Representative lithium signal	Representative TDS signal	Why it matters
Appalachian	Marcellus subsets reach 127-205 mg/L Li	Often >100,000 mg/L TDS	High Li, chemically variable
Permian	Wolfcamp average near 14 ppm Li	<25-140 g/L TDS range	High water volume, lower Li grade
Williston	Brines can exceed 100 mg/L Li	Bakken/Three Forks near 308 g/L TDS	Salinity can dominate pretreatment
Smackover	Observed Li about 1-477 mg/L	156,000-340,000 mg/L TDS	Premium but thermodynamically hard

The case study starts with basin screening because the same extraction train will not behave the same across these feeds.

1.2 Why Smackover Is The Flagship Case

Why this location makes sense

- Southern Arkansas Smackover combines high Li with very high salinity, so the chemistry is both valuable and hard.
- Smackover and Marcellus are the more favorable U.S. plays on major-metals-to-lithium and divalent-to-lithium ratios in the cited screening work.
- Arkansas already has commercial bromine and brine-handling infrastructure.
- That combination connects chemistry, location, and process deployment.

Signal	Smackover basis
Li range	observed wells span about 1-477 mg/L
TDS range	observed samples span 156,000-340,000 mg/L
Median TDS	current observation-set median is 305,000 mg/L
Infrastructure	existing Arkansas bromine/brine operations

1.3 Smackover Feed Basis

Case	Row	Li (mg/L)	TDS (mg/L)	Na/Ca/Mg (mg/L)	Role
Low observed	SH-2	11.7	156,000	37,100 / 13,900 / 2,250	Lower source bound
Base observed	MS-2	168	305,000	64,100 / 36,900 / 3,310	Median-TDS base proxy
High observed	BR-2	252	340,000	70,800 / 39,900 / 3,130	High-TDS/high-Li proxy

These are source rows from the USGS southern-Arkansas Smackover data release after duplicate and blank rows are excluded. The separate 100/200/400 mg/L Li rows are sensitivity cases, not direct well samples.

1.4 Selected Base Brine

Field	Value
Source row	MS-2 / MSPU 4-W1
Formation	Smackover
Li	168 mg/L
TDS	305,000 mg/L
Na / Ca / Mg	64,100 / 36,900 / 3,310 mg/L
Cl / Br	174,000 / 2,700 mg/L

The feed is Smackover-specific. The extraction chemistry is Shan/Gando HBTA/TOPO. The model couples a source-backed brine to a literature-backed non-ionic extraction chemistry without claiming a fully calibrated Arkansas extraction experiment.

1.5 Critical Minerals And REE Boundary

Group	What the Smackover source table supports	How to present it
Reported trace fields	Fe, Mn, Zn, Al, As, Be, Cd, Co, Cu, Cr, Hg, Mo, Ni, Pb, U, V	Use as trace/pretreatment context, with censoring caveats
REE	not_reported in the source table	Do not infer; make this a future-data need

1.6 Variables The Case Must Vary

Axis	Why it matters	Process consequence
Li concentration	Sets value and concentration driving force	Changes recovery target and stage count
TDS / NaCl load	Changes ionic strength and activity behavior	Moves distribution behavior and surrogate validity
Competing ions	Mg, Ca, Sr, Ba, and Na affect selectivity	Changes pretreatment and solvent burden
Solvent / extractant chemistry	Determines whether selective transfer is plausible	Changes (D _i), selectivity, and loaded-solvent composition
O/A ratio	Directly affects stage transfer	Becomes a design and optimization variable

2. ePC-SAFT Thermodynamic Role

2.1 The Model Gap

What weak placeholders can do

- Reproduce a narrow demonstration point
- Provide a quick stage recovery factor
- Run fast inside a flowsheet
- Help debug material balances

What they cannot defend

- Basin-to-basin extrapolation
- Hypersaline electrolyte behavior
- Solvent and O/A sensitivity
- Ion-specific distribution behavior
- Trust-region bounds for optimization

The process model needs a physical equilibrium layer before the surrogate or optimizer is meaningful.

2.2 What ePC-SAFT Adds

Modeling need	ePC-SAFT contribution	Why it is novel for this case
Hypersaline brine	Electrolyte activity and phase-equilibrium structure	Salinity becomes a modeled variable, not a correction factor
Aqueous/organic split	Phase compositions and phase fractions	The extraction stage gets a thermodynamic target
Ion partitioning	Distribution ratios and selectivity metrics	Li movement can be separated from bulk salt movement
Sensitivity surfaces	Response to TDS, O/A, solvent, and feed composition	Surrogates can be trained inside a meaningful region
Diagnostics	Stability, convergence, and phase-distance checks	Failed or collapsed points do not silently become process data

2.3 Rezaee 2026 As A Parameter-Fitting Smoke Test

What Rezaee gives	How this deck should use it
DES/TOPO lithium extraction modeling with tabulated PC-SAFT-style parameters	Package regression and equilibrium smoke test
Organic phase modeled with PC-SAFT	Do not cite as full ePC-SAFT organic-phase closure
Aqueous electrolyte phase modeled with ePC-SAFT	Useful support for the aqueous-brine side of the bridge
Fitted organic interactions and product pseudo-species	Useful parameterization pattern, not Shan/Gando chemistry

Rezaee is a method lead and runtime test, not the final HBTA/TOPO model.

2.4 Solvent Candidate Scorecard

Priority	System	Role	Status
1	HBTA + TOPO + sulfonated kerosene	Flagship	Field-water evidence, source-regressed Li/Na stage model, missing full multication parameter payload
2	D2EHDTPA + BuPhen + octanol + n-dodecane	Best non-HBTA backup	Strong multication selectivity, blocked by missing parameter set
3	TBAC + decanoic-acid DES + TOPO	Parameter-regression pilot	Rezaee smoke test runs; not the flagship chemistry
4	TBP + FeCl3/HCl	Mechanistic backup	Useful high-Mg comparison, process-burden and parameter gaps

2.5 Limitation Of Fixed-Coefficient Crossflow

Placeholder limitation

- The legacy Jang-style crossflow placeholder reaches only 39.96% cumulative Li extraction after ten contacts.
- Sodium still reaches 38.05% cumulative extraction in that same chain.
- It is a limitation baseline, not a flagship produced-water case.

Why it was not enough

- fixed surrogate chemistry
- weak Li-over-Na separation
- no bridge to trust-region sampling or ALAMO

3. Selective Extraction Model

3.1 One-Stage Selective Split

MS-2 O/A = 1 stage result

- Li extraction: 47.2846%
- Na extraction: 0.0131%
- ($D_{\text{Li}} = 0.8970$)
- ($S_{\text{Li/Na}} = 6840.1$)
- Raffinate Li: 88.56 mg/L
- Activity correction: ePC-SAFT aqueous gamma

Source-regressed Li/Na stage solve; not a full multication reactive LLE flash.

What changed from the old wrapper

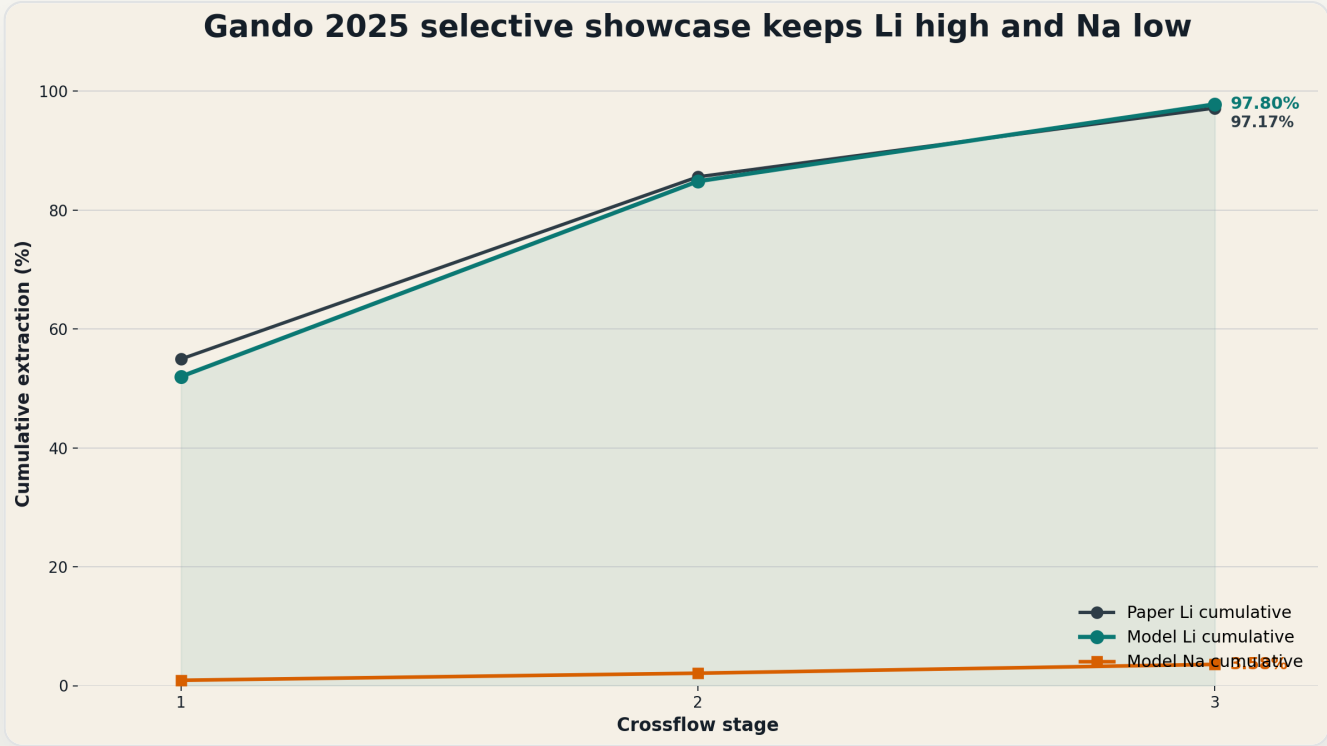
- The chemistry now carries the literature-backed 2 HBTA : 1 TOPO : 1 Li complex stoichiometry.
- The aqueous Li/Na activity correction is evaluated by ePC-SAFT when available.
- The old selective wrapper remains only as historical support and fallback context.
- The current stage-output table carries extraction, distribution, selectivity, and validity labels.

3.2 Li/Na Regression Closure

Anchor	Zotero key	Target	Prediction	Residual
Stage 1 Li cumulative extraction	JUNBXVTI	54.95%	54.8882%	-0.0618 pp
Stage 2 Li cumulative extraction	JUNBXVTI	85.60%	85.7224%	+0.1224 pp
Stage 3 Li cumulative extraction	JUNBXVTI	97.17%	97.0385%	-0.1315 pp
Li/Na selectivity lower-bound anchor	9LJWDC7E	2100	2100.008	+0.008

This is the current predictive closure: Li/Na staged transfer from one source-regressed parameter payload. Ca, Mg, Ba, Sr, diluent, and complex parameters remain explicit limits.

3.3 Multi-Stage Performance



Three-stage reactive-stage chain

- Stage 1 cumulative Li extraction: 47.2846%
- Stage 2 cumulative Li extraction: 89.5753%
- Stage 3 model Li extraction: near-total
- Stage 3 cumulative Na extraction: 0.1160%
- Costing recovery is capped at the 97.17% literature anchor

3.4 Thermodynamic Insight

The old question

Can this solvent extract lithium from this brine?

The stronger question

Where should this chemistry work, how does the answer change with basin chemistry, and what stage variables should enter a process model?

Why ePC-SAFT changes the question

- It turns feed chemistry into equilibrium states.
- It lets salt load and O/A ratio become computable sensitivities.
- It provides distribution and selectivity variables, not only recovery percentages.

4. PrOMMiS / IDAES Process Bridge

4.1 From Equilibrium To Transfer Variables

Process-model need	ePC-SAFT-derived quantity	Use in PrOMMiS / IDAES
Feed state	(z _i), TDS, Li/Na/Mg/Ca/Sr/Ba/Cl basis, temperature	Basin-specific solvent-extraction inlet
Phase split	aqueous and organic compositions, phase fraction	Equilibrium target or closure for a stage
Transfer strength	(D _i = C _{i,org} /C _{i,aq})	Distribution behavior in material balances
Selectivity	(S _{Li/Na}), later (S _{Li/Mg}), (S _{Li/Ca})	Separation constraints and solvent screening

PrOMMiS and IDAES do not just need a lithium recovery number. They need a defensible map from produced-water chemistry to stage transfer variables.

4.2 Transfer-Variable Contract

For process integration, the ePC-SAFT output should be treated as the following structured contract:

Variable	Definition	Why PrOMMiS / IDAES needs it
(_i)	stage removal (= ($C_{\{i,in\}} - C_{\{i,out\}}$) / $C_{\{i,in\}}$)	Links chemistry to process-wide recovery objectives
(D _i)	($C_{\{i,org\}} / C_{\{i,aq\}}$)	Controls staged transfer in material balance equations
(S _{Li/Na}), (S _{Li/Mg}), (S _{Li/Ca})	Selectivity ratios from stage/phase outputs	Enables solvent and design comparison without refitting fixed coefficients
Validity flags	residuals, phase flags, domain bounds	Prevents brittle points from entering surrogate or optimizer datasets

This keeps the story concrete: ePC-SAFT computes the transfer map; PrOMMiS/IDAES consumes it.

4.3 Why This Belongs In PrOMMiS / IDAES

1. Offline engine Run ePC-SAFT outside the flowsheet to generate equilibrium tables over feed chemistry, O/A ratio, and solvent conditions.

2. Surrogate layer Fit ALAMO or another surrogate inside a documented trust region so optimization is fast and bounded.

3. External function Expose high-fidelity calls for studies where direct thermodynamic solves are worth the cost.

4. Process model Use PrOMMiS/IDAES for staged extraction, recycle, stripping, costing hooks, and optimization.

4.4 Formal IDAES Costing Scaffold

Costing block	Current basis	Needed before formal TEA
Feed and annual basis	Smackover base feed; 100/1000/10000 m3/day scenarios	approved flowrate and uptime
Pretreatment	IDAES/Pyomo scenario expression from model inputs	removal target, reagent use, waste handling
Extraction/contacting	three-stage HBTA/TOPO reactive-stage solve	contactor sizing, residence time, solvent inventory
Solvent makeup/loss	low/base/high placeholder per m3 feed	HBTA/TOPO/diluent prices and loss rate
Concentration and Li2CO3 precipitation	product-purity anchor 99.28%; recovery capped at 97.17%	evaporation duty, carbonate reagent, product spec

The formal IDAES/Pyomo layer computes production, revenue, annual operating cost, annualized contactor CAPEX, breakeven Li2CO3 price, and net-before-tax scenario values.

4.5 Implementation Requirements

Add ePC-SAFT as the thermodynamic backbone

- property/equilibrium service for electrolyte LLE
- documented species and feed basis
- diagnostics carried with every result
- reproducible dataset generation

Use PrOMMiS/IDAES for the process system

- solvent-extraction stage balances
- multi-stage contactors
- surrogate and external-function interfaces
- flowsheet optimization and costing hooks

The sell is simple: ePC-SAFT supplies the equilibrium evidence; PrOMMiS/IDAES turns that evidence into a process model.

4.6 Remaining Engineering Work

Priority	Work item	Why it matters
1	Extract Zhang 2017 distribution, O/A, temperature, and capacity data	Improves identifiability of the Li/Na transfer law
2	Add explicit HBTA, TOPO, diluent, and Li-complex parameter regression	Moves from source-regressed transfer to reactive ePC-SAFT closure
3	Build a PrOMMiS MSContactor validation case	Tests convergence and material-balance consistency under staged operation
4	Replace Class-5 costing assumptions with engineering ranges	Improves economic credibility before TEA-style interpretation

5. Backup: Sensitivity Expansion

5.1 Next-Step Sensitivity Expansion

- Re-run the selective case across basin-style feed envelopes:
 - Appalachian high-Li case
 - Permian volume-driven low-grade case
 - Williston high-TDS case
 - Smackover premium-brine case
- Keep the next design-of-experiments table in Markdown so trust-region bounds and surrogate regions are explicit.
- Add a PrOMMiS/IDAES interface table for every generated variable: units, species basis, source basis, and validity limits.
- Keep the narrative in one place: screening, thermodynamics, selective result, and process-design interface.